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Statistical Learning II

Lecture 5 - Least squares (continued)

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Least-squares regression

Let $\mathcal{D} = \{(x_i, y_i) \in \mathbb{R}^d \times \mathbb{R} : i = 1, \dots, n\}$ denote the training data.

Ordinary least-squares (OLS) regression is defined as:

$$\min_{\boldsymbol{\theta} \in \mathbb{R}^d} \hat{\mathcal{R}}_n(\boldsymbol{\theta}) := \frac{1}{2n} ||\mathbf{y} - \mathbf{X}\boldsymbol{\theta}||_2^2$$

Where we have defined the data matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ and label vector $\mathbf{y} \in \mathbb{R}^n$:

$$\mathbf{X} = \begin{bmatrix} - & \mathbf{x}_1 & - \\ - & \mathbf{x}_2 & - \\ & \vdots & \\ - & \mathbf{x}_n & - \end{bmatrix} \in \mathbb{R}^{n \times d} \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

Convexity of OLS

$$\hat{\mathcal{R}}_n(\boldsymbol{\theta}) := \frac{1}{2n} ||\mathbf{y} - \mathbf{X}\boldsymbol{\theta}||_2^2$$

- Gradient: $\nabla_{\boldsymbol{\theta}} \hat{\mathcal{R}}_n = -\frac{1}{n} \mathbf{X}^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\theta}) \in \mathbb{R}^d$
- Hessian: $\nabla_{\boldsymbol{\theta}}^2 \hat{\mathcal{R}}_n = \frac{1}{n} \mathbf{X}^\top \mathbf{X} \in \mathbb{R}^{d \times d} \quad (:= \hat{\boldsymbol{\Sigma}}_n)$

Since $\mathbf{X}^\top \mathbf{X} \succeq 0$, $\hat{\mathcal{R}}_n$ is **convex** over $\mathbb{R}^d$. This implies that any minimum of $\hat{\mathcal{R}}_n$ is a global minimum.

For $n \geq d$, $\hat{\mathcal{R}}_n$ is **strictly convex** if and only if $\text{rank}(\mathbf{X}^\top \mathbf{X}) = d$. This implies that $\hat{\mathcal{R}}_n$ can have at most one global minimum.

Explicit solution

- Gradient: $\nabla_{\boldsymbol{\theta}} \hat{\mathcal{R}}_n = -\frac{1}{n} \mathbf{X}^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\theta}) \in \mathbb{R}^d$

If it exists, a minima must satisfy:

$$\nabla_{\boldsymbol{\theta}} \hat{\mathcal{R}}_n \stackrel{!}{=} 0$$

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$$\nabla_{\boldsymbol{\theta}} \hat{\mathcal{R}}_n \stackrel{!}{=} 0 \quad \Leftrightarrow \quad \mathbf{X}^\top \mathbf{X} \boldsymbol{\theta} = \mathbf{X}^\top \mathbf{y}$$

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If $\mathbf{X}^\top \mathbf{X}$ is invertible, unique solution:

$$\hat{\boldsymbol{\theta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

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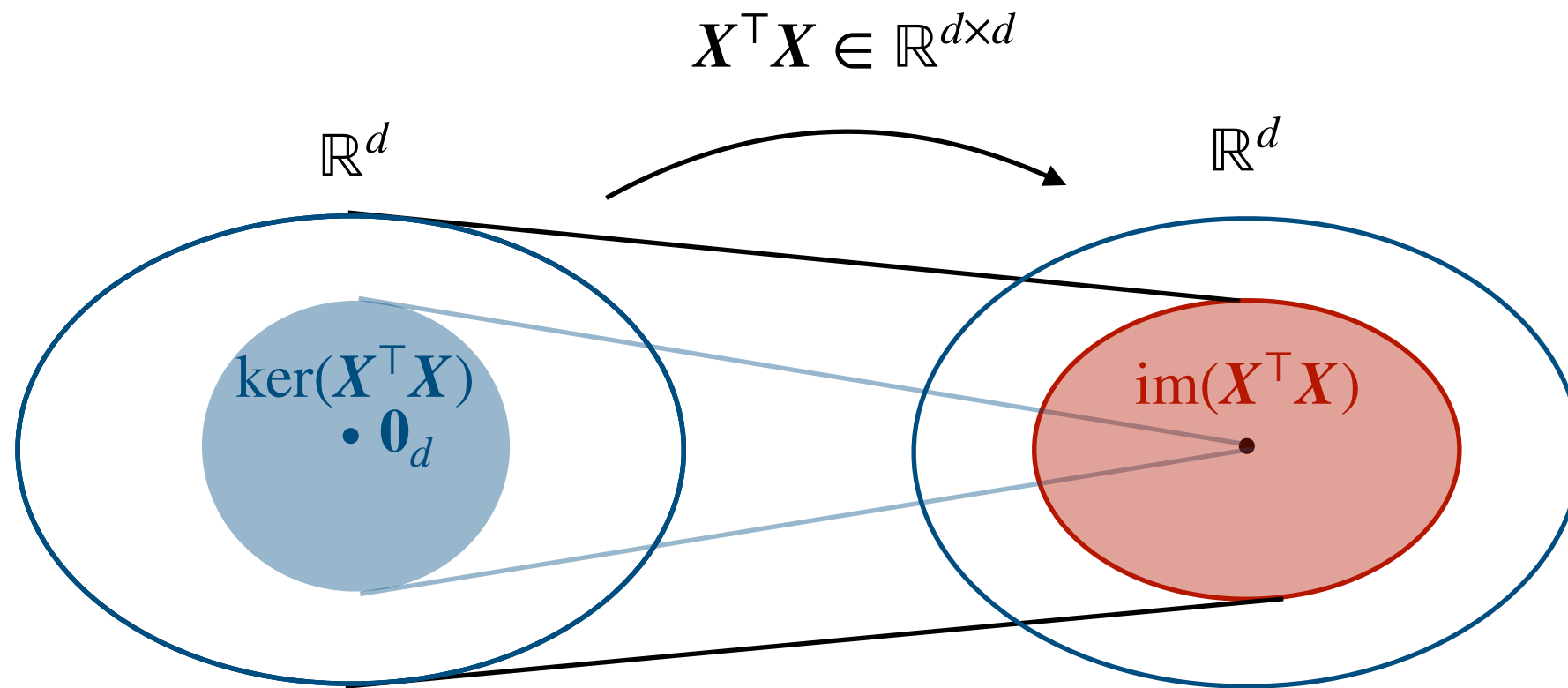


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But what if $\mathbf{X}^\top \mathbf{X}$ is **not invertible**? For example, if $\text{rank}(\mathbf{X}) = n < d$?

Two scenarios

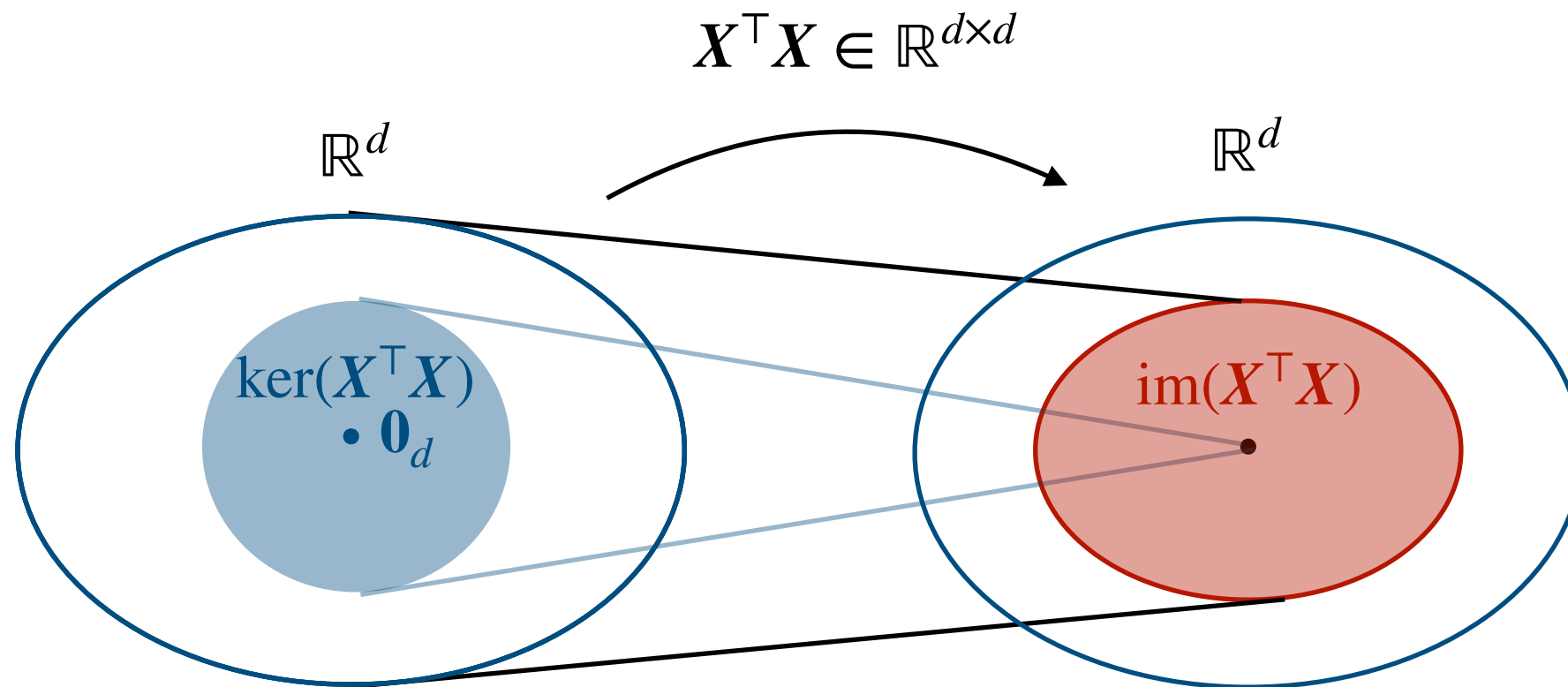
Focus on case $\text{rank}(X) = n < d$ (i.e. X is full-rank)



Note $\text{rank}(X) = \text{rank}(X^T X) = \text{rank}(X X^T)$

Two scenarios

Focus on case $\text{rank}(X) = n < d$ (i.e. X is full-rank)



Note $\text{rank}(X) = \text{rank}(X^T X) = \text{rank}(XX^T)$

All solutions of $X^T X \boldsymbol{\theta} = X^T \mathbf{y}$ can be written as:

$$\hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}}_0 + \mathbf{k}$$

Where: $\mathbf{k} \in \ker(X^T X) \simeq \mathbb{R}^{d-n}$ and $\hat{\boldsymbol{\theta}}_0 \in \text{im}(X^T X) \simeq \mathbb{R}^n$

Explicit solution

For $\text{rank}(\mathbf{X}) = n < d$, a particular solution of $\mathbf{X}^\top \mathbf{X} \boldsymbol{\theta} = \mathbf{X}^\top \mathbf{y}$ is:

$$\hat{\boldsymbol{\theta}}_0 = \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top)^{-1} \mathbf{y} \quad (\text{Check this!})$$

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Together, in the full-rank case $\text{rank}(\mathbf{X}) = \min(n, d)$ solution is:

$$\hat{\boldsymbol{\theta}} = \begin{cases} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} & \text{for } n \geq d \\ \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top)^{-1} \mathbf{y} + \mathbf{k} & \text{for } n < d \end{cases}$$

For any $\mathbf{k} \in \ker(\mathbf{X}^\top \mathbf{X})$.

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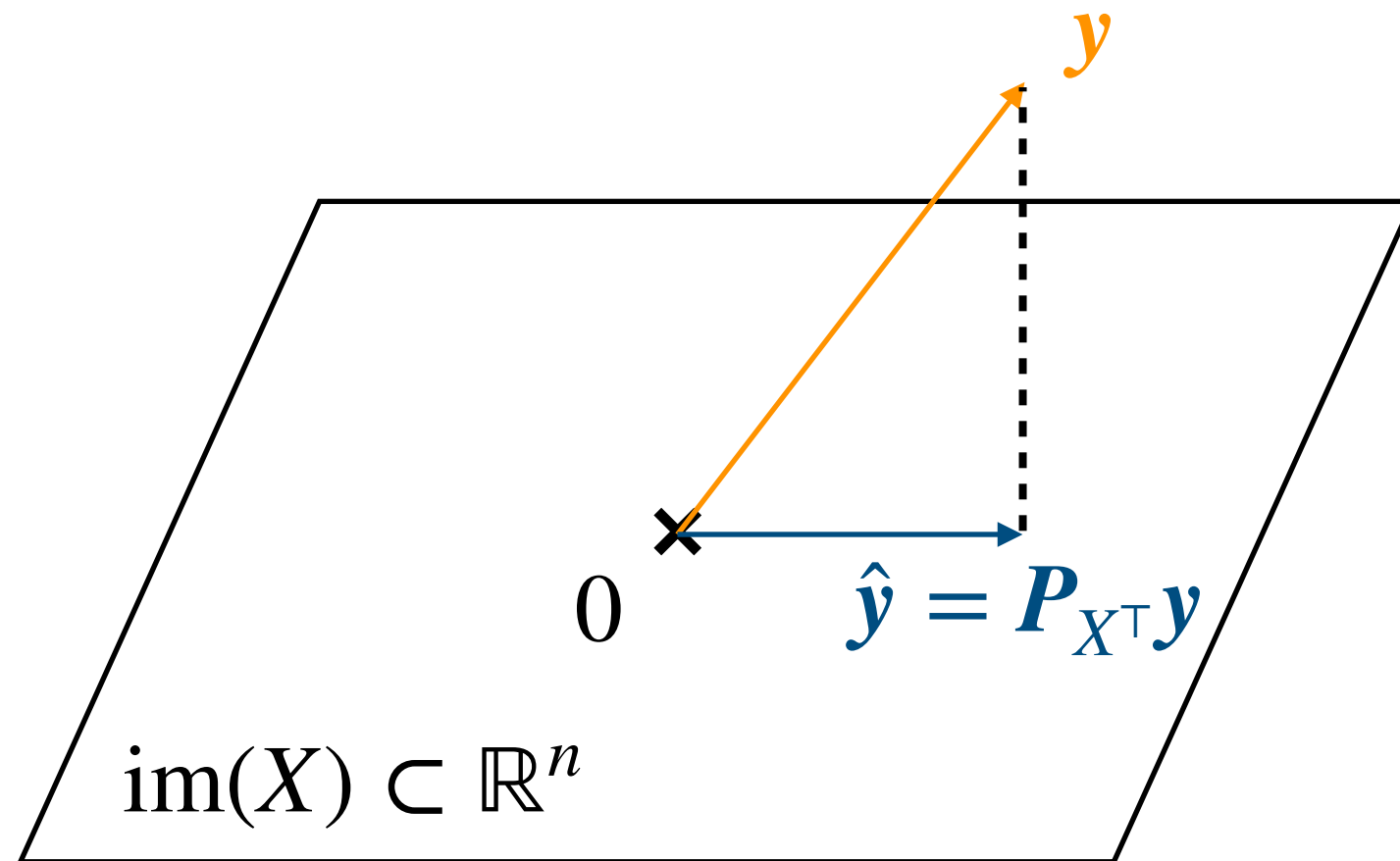
In particular, for $\mathbf{k} = \mathbf{0} \in \ker(\mathbf{X}^\top \mathbf{X})$ this is the **Moore-Penrose inverse**:

$$\hat{\boldsymbol{\theta}}_{\text{OLS}} = \mathbf{X}^+ \mathbf{y}$$

Geometrical interpretation

This gives a natural interpretation of the OLS predictor as an orthogonal projection of the labels in the row space of X :

$$\hat{\theta}_{OLS} = X^+ y \quad \Rightarrow \quad \hat{y}_{OLS} = X \hat{\theta}_{OLS} = X X^+ y$$



$$\min_{z \in \text{im}(X)} \|y - z\|_2^2$$

OLS as least norm solution

Assume $\text{rank}(\mathbf{X}) = n < d$. Then, OLS admits the following interpretation as the minimum ℓ_2 -norm solution:

$$\begin{aligned} \hat{\boldsymbol{\theta}}_{OLS} = \operatorname{argmin}_{\boldsymbol{\theta} \in \mathbb{R}^d} ||\boldsymbol{\theta}||_2 \\ \text{subject to} \quad \mathbf{X}\boldsymbol{\theta} = \mathbf{y} \end{aligned}$$

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